HarvardX

Professional Certificate

CS50's Computer Science

COURSE	PREREQUISITES	CONTENT	DURATION
CS50's Understanding Technology	Language: English None	1. Internet 2. Multimedia 3. security 4. web development 5. programming	Weeks: 6 Hours: 2 - 6 Self-Placed
CS50's Introduction to Computer Science	Language: English None	1. computer science and programming 2. think algorithmically and solve problems efficiently 3. concepts - abstraction, algorithms, data structures 4. concepts - encapsulation, resource management, security 5. concepts - software engineering and web development 6. familiarity with - C, Python, SQL, and JS, CSS and HTML 7. vibrant community of like-minded learners from all levels	Weeks: 12 Hours: 6 - 18 Self-Placed
CS50's web programming with python and js	Language: English CS50 / programming	1. HTML, CSS 2. Git 3. Python 4. Django 5. SQL, Models, and Migrations 6. JavaScript 7. User Interfaces 8. Testing, CI/CD 9. Scalability and Security	Weeks: 12 Hours: 6 – 9 Self-Placed
CS50's Mobile App with React Native	Language: English CS50 / HTM, CSS, JS	1. JavaScript 2. ES6 3. React, JSX 4. Components, Props, State, Style 5. Components, Views, User Input 6. Debugging 7. Data 8. Navigation 9. Expo Components 10. Redux	Weeks: 13 Hours: 6 – 9 Self-Placed

		11. Performance 12. Shipping, Testing	
CS50's Introduction to AI with Python	Language: English CS50 / python	1. Graph search algorithms 2. Adversarial search 3. Knowledge representation 4. Logical inference 5. Probability theory 6. Bayesian networks 7. Markov models 8. Constraint satisfaction 9. Machine learning 10. Reinforcement learning 11. Neural networks 12. Natural language processing	Weeks: 7 Hours: 10 – 30 Self-Placed
CS50's game Development	Language: English CS50 / programming	1. Principles of 2d and 3d graphics 2. Animation, Sound, Collision detection 3. Frameworks like unity and love 2d 4. Language like – lua and c# 5. Understanding of basics of game design a d development	Weeks: 12 Hours: 6 – 9 Self-Placed
CS50's Computer Science for Lawyers	Language: English None	1. Computational Thinking 2. Programming languages 3. Algorithms, data structures 4. Cryptography 5. Cybersecurity 6. Internet technologies, cloud computing 7. Web programming 8. Database design 9. Cybersecurity, continued 10. Challenges at the intersection of law and technology	Weeks: 10 Hours: 3 – 6 Self-Placed
Cost: 313.20 USD (30,000 BDT) – Introduction to Computer Science + Technology | Web | Mobile | Game | AI | Lawyers			
Professional Certificate Data Science	Language: English	1. Data Science: R Basics 2. Data Science: Visualization 3. Data Science: Probability 4. Data Science: Inference and Modeling 5. Data Science: Productivity Tools 6. Data Science: Wrangling 7. Data Science: Linear Regression 8. Data Science: Machine Learning	1 Year 5 Months Hours: 2 – 3 Self-Placed Cost: 792 USD